TECHNISCHE UNIVERSITÄT MÜNCHEN

# Quantum Chemistry Laboratory

**Prof. N. Rösch**

# Full Solution of Exercise 12

| Method/Basis Set | Atomization Enthalpies, kJ/mol |
|---|---|
| HF/6-31G(d) | 1002 |
| MP2/6-31G(d) | 1409 |
| LSDA/6-31G(d) | 1763 |
| Becke3LYP/6-31G(d) | 1501 |
| G1 | 1513 |
| Experimental Value | 1519 |

## Answers to the Questions:

a. G1

b. Becke3LYP

c. Zero point energy ($CH_2O$) = 70 kJ/mol

d. $\Delta H = c_p \cdot \Delta T = 5/2\ R \cdot \Delta T = 6.2$ kJ/mol; $\Delta H_{B3LYP} = 10.0$ kJ/mol

Difference is due to internal degrees of freedom in $CH_2O$ (no monatomic gas).

e. $E_{MP2(frozen\ core)} - E_{MP2(full)} = 0.00196$ hartree

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[General help]



[To overview]



[Compute]



[Contact the course supervisor]

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